We consider the following sequence of forces

$$f_h(x') = h^{\gamma+1}[x_1 e_3 + h x_2 e_2].$$

Note that the sequence $f_h$ is mean-free by simmetry. Then, with some simple computation one has that

$$F_h(RU_1) = \frac{h^{\gamma+1}}{12}(R_{31} + hR_{22}),$$
$$F_h(RU_2) = \frac{h^{\gamma+1}}{12}(R_{31} + 2hR_{22}).$$

It follows that $\mathcal{R}_h^1 = \mathcal{R}_h^2$ are singletons given by the matrix

$$\begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix},$$

and that $\Lambda_h = \{2\}$. The limit force $f$ is given by $f(x') = x_1 e_3$. Hence,

$$F_h(RU_1) = F(RU_2) = \frac{1}{12} R_{31},$$

Thus,

$$\mathcal{R}^1 = \mathcal{R}^2 = \{R \in \mathrm{SO}(3) : R_{31} = 1\},$$

and $\dim \mathcal{R}^1 = \dim \mathcal{R}^2 = 1$. Moreover, $\Lambda = \{1, 2\}$.

Under the assumptions (F1)–(F2), the following result holds.

**Theorem 1.7.** *Let $\alpha > 2$ and $q > 1$. Assume (F1)–(F2). Suppose that $(y_h) \subset W^{1,2}(\Omega; \mathbb{R}^3)$ is a sequence of deformations that are quasi-minimizers for $J_h^\alpha$, i.e.,*

$$\limsup_{h\to 0} (J_h^\alpha(y_h) - \inf J_h^\alpha) = 0\,.$$

*Then, $E_h^\alpha(y_h) \le C$ for every $h > 0$ and there exist $(\bar{R}_h) \subset \mathrm{SO}(3)$, $(c_h) \subset \mathbb{R}$, $u \in W^{1,2}(S; \mathbb{R}^2)$, $v \in W^{2,2}(S)$, and $j \in \{1, \dots, l\}$ such that $j \in \Lambda$ and, up to subsequences,*

*(i) $\bar{R}_h \to \bar{R}$ with $\bar{R}^T U_j \in \mathcal{M}$,*

*(ii) $u_h \rightharpoonup u$ in $W^{1,2}(\Omega; \mathbb{R}^2)$,*

*(iii) $v_h \to v$ in $W^{1,2}(\Omega)$,*

*where $u_h, v_h$ are defined as in (4)–(5) for $\tilde{y}_h = \bar{R}_h y_h + c_h$. In addition, there is $W \in \mathbb{R}^{3\times 3}_{\mathrm{skew}}$ such that $|W| = 1$, $\bar{R}^T W \in N\mathcal{R}^j_{\bar{R}^T}$, and*

$$h^{\frac{1}{2}(1-\gamma)}(\bar{R}_h^T - P_h^j(\bar{R}_h)) \to \beta \bar{R}^T W\,,$$

*for some $\beta \ge 0$. Lastly,*

*(a) if $2 < \alpha < 4$ and $S$ is a simply connected set that satisfies (2), then $(j, v, \bar{R}^T, \beta W)$ minimizes the functional*

$$J_j^{\mathrm{CVK}}(v, R, W) = E_j^{\mathrm{CVK}}(v) - \int_S f \cdot RU_j^{-1} e_3 v\, dx - F(RW^2 U_j)$$

*over all the admissible quadruplets $(j, v, R, W)$, i.e., $j \in \Lambda$ and $(v, R, W) \in \mathcal{C} \times \mathcal{R}^j \times \mathbb{R}^{3\times 3}_{\mathrm{skew}}$ is such that $W \in N\mathcal{R}^j_{R^T}$, where $\mathcal{C}$ is the set*

$$\mathcal{C} = \left\{ v \in W^{2,2}(S) \colon \exists\, u \in W^{1,2}(S; \mathbb{R}^2) \text{ s.t. } \nabla u^T + \nabla u + |U_j^{-1} e_3| \nabla v \otimes \nabla v = 0 \right\}\,,$$

*(b) if $\alpha = 4$, then $(j, u, v, \bar{R}^T, \beta W)$ minimizes the functional*

$$J_j^{\mathrm{VK}}(u, v, R, W) = E_j^{\mathrm{VK}}(u, v) - \int_S f \cdot RU_j^{-1} e_3 v \, dx - F(RW^2 U_j)$$

*over all the admissible quintuplet $(j, u, v, R, W)$, i.e., $j \in \Lambda$ and $(u, v, R, W) \in W^{1,2}(S; \mathbb{R}^2) \times W^{2,2}(S) \times \mathcal{R}^j \times \mathbb{R}^{3\times 3}_{\mathrm{skew}}$ is such that $W \in N\mathcal{R}^j_{R^T}$,*

*(c) if $\alpha > 4$, then $(j, u, v, \bar{R}^T, \beta W)$ minimizes the functional*

$$J_j^{\mathrm{LVK}}(u, v, R, W) = E_j^{\mathrm{LVK}}(u, v) - \int_S f \cdot RU_j^{-1} e_3 v \, dx - F(RW^2 U_j)$$

*over all the admissible quintuplets $(j, u, v, R, W)$, i.e., $j \in \Lambda$ and $(u, v, R, W) \in W^{1,2}(S; \mathbb{R}^2) \times W^{2,2}(S) \times \mathcal{R}^j \times \mathbb{R}^{3\times 3}_{\mathrm{skew}}$ is such that $W \in N\mathcal{R}^j_{R^T}$.*

**Remark 1.8.** Given that the scaling of the forces is of order $h^{\gamma+1}$, we expect the action of the load on the in-plane displacement to be negligible. This is indeed the case and the limiting forcing term acts only on the out-of-plane displacement. The additional term $F(RW^2U_j)$ can be interpreted (see also [21]) as an elastic cost of fluctuations of the reference configuration from the optimal rotations.

From a minimization point of view, since the term $F(RW^2U_j)$ is always non-positive (see Section 5), it is clear that the optimal choice is $W = 0$. In particular, in Theorem 1.7 we actually have $\beta = 0$. Similarly, since

$$E_j^{\mathrm{LVK}}(0, v) \leq E_j^{\mathrm{LVK}}(u, v) \quad \forall\, u \in W^{1,2}(S; \mathbb{R}^2) \quad \forall\, j = 1, \dots, l\,,$$

in Theorem 1.7–(c) we also have $u = 0$.

We provide now an example of rank-one connected double-well structure for which different applied forces result in different preferred reference configurations.

**Example 1.9.** Let $S = (-\frac{1}{2}, \frac{1}{2})^2$ and consider

$$U_1 = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad U_2 = \begin{pmatrix} 2 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}.$$

For $a$, $b$, $c \geq 0$, consider the sequences of loads

$$f_h(x') = h^{\gamma+1}(ax_1e_1 + bx_2e_2 + cx_1e_3).$$

The limit force $f(x') = ax_1e_1 + bx_2e_2 + cx_1e_3$ is pulling the mid-plane $S$ along fibers parallel to $e_1$ and $e_2$, while twisting it in the out-of-plane direction. With some simple computation we get

$$\begin{aligned} F(RU_1) &= \frac{1}{12}(4aR_{11} + bR_{22} + cR_{31}), \\ F(RU_2) &= \frac{1}{12}[a(2R_{11} + R_{13}) + bR_{22} + c(2R_{31} + R_{33})]. \end{aligned}$$

Note that, if $a = 0$ and $b$, $c > 0$ (i.e., $f$ is pulling the mid-plane $S$ in the $e_2$ direction only), then one has that

$$F(RU_1) \leq \frac{1}{12}(b + c) < F(\bar{R}U_2), \qquad \forall\, R \in \mathrm{SO}(3),$$

where

$$\bar{R} = \begin{pmatrix} -\frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} \\ 0 & 1 & 0 \\ \frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} \end{pmatrix}.$$

In particular, $\Lambda = \{2\}$ and the only admissible well at the limit is $\mathrm{SO}(3)U_2$. However, if $a > 0$ and $b = c = 0$, that is $f$ is tensing $S$ along $e_1$ without twisting the mid-plane, we have

$$F(RU_2) = \frac{a}{12}(2R_{11} + R_{13}) \leq \frac{a}{4} < \frac{a}{3} = F(U_1), \qquad \forall\, R \in \mathrm{SO}(3)\,.$$

Thus, $\Lambda = \{1\}$ and the only admissible reference configuration is $\mathrm{SO}(3)U_1$.

## 2 Preliminary results

In this section we give a few preliminary results regarding the energy $W$. These properties are well known in the case of a single well elastic energy. Firstly, we explore the invariance properties of the second derivative $Q_j$ that follow from frame indifference (W3). Lastly, we prove that (W4) implies some coercivity for $Q_j$.

**Lemma 2.1.** *The following equality holds true for every $j = 1, \dots, l$:*

$$Q_j(A) = Q_j(\mathrm{sym}(AU_j^{-1})U_j) = Q_j(U_j^{-1}\,\mathrm{sym}(U_jA)) \quad \forall\, A \in \mathbb{R}^{3\times 3}\,.$$

*Proof.* Define $\tilde{W}(F) = W(FU_j)$. By the properties (W1)–(W3) we deduce that $\tilde{W}$ is frame indifferent, it is $C^2$ in a neighborhood of $\mathrm{SO}(3)$, and it is minimized on $\mathrm{SO}(3)$. It is well known that any map satisfying these properties has symmetric second derivative at the identity, namely:

$$D^2\tilde{W}(\mathrm{Id})[A]^2 = D^2\tilde{W}(\mathrm{Id})[\mathrm{sym}(A)]^2\,.$$

By some simple computation we get $D^2\tilde{W}(\mathrm{Id})[A]^2 = Q_j(AU_j)$. Hence,

$$Q_j(A) = Q_j(AU_j^{-1}U_j) = Q_j(\mathrm{sym}(AU_j^{-1})U_j)\,.$$

To conclude we observe that by definition of symmetric part and the symmetry of $U_j$ we have

$$\begin{aligned} Q_j(\mathrm{sym}(AU_j^{-1})U_j) &= Q_j(\mathrm{sym}(U_j^{-1}U_jAU_j^{-1})U_j) \\ &= Q_j(U_j^{-1}\,\mathrm{sym}(U_jA))\,. \end{aligned}$$

■

**Lemma 2.2.** *There exists $\lambda > 0$ such that for every $j = 1, \dots, l$ and every $A \in \mathbb{R}^{3\times 3}$*

$$Q_j(U_j^{-1}\,\mathrm{sym}(A)) \geq \lambda|\,\mathrm{sym}(A)|^2\,.$$

*Proof.* By (W4) we have, for $\varepsilon \ll 1$

$$W(U_j + \varepsilon U_j^{-1}A) \geq C\,\mathrm{dist}^2(U_j + \varepsilon U_j^{-1}A, K) = C\,\mathrm{dist}^2(U_j + \varepsilon U_j^{-1}A, K_j)\,.$$

By the Taylor expansion of $W$ at $U_j$ and by Lemma 2.1 we get

$$\begin{aligned} \varepsilon^2 Q_j(U_j^{-1}\,\mathrm{sym}(A)) + o(\varepsilon^2) &\geq C\,\mathrm{dist}^2(U_j + \varepsilon U_j^{-1}A, K_j) \\ &\geq C\,\mathrm{dist}^2(\mathrm{Id} + \varepsilon U_j^{-1}AU_j^{-1}, \mathrm{SO}(3)) \\ &= C\varepsilon^2|\,\mathrm{sym}(U_j^{-1}AU_j^{-1})|^2 + o(\varepsilon^2) \\ &= C\varepsilon^2|U_j^{-1}\,\mathrm{sym}(A)U_j^{-1}|^2 + o(\varepsilon^2) \\ &\geq C\varepsilon^2|\,\mathrm{sym}(A)|^2 + o(\varepsilon^2)\,. \end{aligned}$$

Dividing by $\varepsilon^2$ and passing to the limit as $\varepsilon \to 0$ we conclude. ■